

AMTH Thesis: Conrad Lee

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1 Introduction

This undergraduate senior thesis project seeks to investigate whether the Ratings Percentage Index (RPI) systematically over-values teams from stronger Division 1 conferences as a metric of comparison for the NCAA Division 1 Men's College Soccer Championship. The project will begin with a formal mathematical analysis of the RPI formula as used by NCAA Men's Soccer and the intuition behind its potential bias towards teams from stronger Division 1 conferences in contrast to a more standard Bradley-Terry pairwise comparison model. Then, NCAA regular-season games will be simulated to create synthetic RPI and Bradley-Terry rankings and to compare the relative under-performance by teams from stronger conferences across the two pairwise comparison models. After that, the inference methods will be applied to real-life NCAA college soccer data to reveal how simulation patterns hold with real-life results. Lastly, a detailed analysis of findings will provide insights on the value of RPI as a metric for evaluating NCAA men's soccer teams across conferences in the NCAA tournament selection process.

2 Background and Motivation

The National Collegiate Athletic Association (NCAA) is the chief administrator for varsity college athletics in the United States. The body oversees 24 different sports, typically divided between men's and women's teams and across three different divisions. This includes Men's Soccer, also known outside of the United States as Association Football. (This is distinct from American Football, which is sanctioned separately by the NCAA). The largest universities compete at the Division I (D1) level in Men's Soccer. As of the 2024 season, there are 212 NCAA Men's Soccer teams at the D1 level, competing against each other in 22 conferences as well as in non-conference matchups. Official NCAA matches are conducted during the fall months, typically from August to December.

Around midway through each season, the NCAA begins to publish its Adjusted Rating Percentage Index (RPI) rankings for the 212 D1 Men's Soccer teams. RPI provides a single metric to evaluate all teams, regardless of conference, based on their record and their strength of schedule. The NCAA updates

these rankings weekly until the NCAA D1 Men's Soccer Championship begins; other third-party online sources provide daily updates by calculating the metric independently.

The NCAA provides a D1 championship for Men's Soccer (as well as for most other sports) as outlined by the annual NCAA Division I Manual, following the conclusion of all regular-season matchups and conference championships. As of the 2024 edition, the Men's Soccer championship is intended "to provide national-level competition among the best eligible student-athletes and teams of member institutions" (316). Selection to the championship also has lucrative financial implications for participating teams (source). The procedure for selection to the championship is specified by the NCAA's 2024-25 DI Men's Soccer Championship Prechampionship Manual. The championship is formatted in a 48-team, single-elimination bracket. The winners of each of the 22 conferences receive automatic qualification. The 26 remaining teams are selected at-large by an 8-member selection committee in a process detailed by the NCAA. RPI plays a key role in this process. The procedure for this process is outlined in the flow chart "NCAA RPI Procedure" and will be elaborated on here.

2.1 Selection Procedure

According to the DI Men's Soccer Championship Prechampionship Manual, selection will be determined on the basis of three criteria, which selection for all NCAA Championships are determined by: "Won-lost record; strength of schedule; and eligibility and availability of student-athletes for NCAA championships" (15). In addition, a team must have a DI won-lost-tied record of .500 or better to be eligible for selection.

The Men's Soccer Committee must also consider other criteria specific to the Men's Soccer format. This includes the Adjusted RPI, which includes the following: - Division I winning percentage with ties calculated as 1/3 of a win (25- Opponents' strength of schedule (50- Opponents' opponents' strength of schedule (25- A bonus/penalty system (15)

In addition, the committee must consider the following criteria, which will eventually be summarized in a table in the Appendix: - Head-to-head competition - Results versus common opponents - Strength and results against nonconference opponents - Result against teams already selected (including automatic qualifiers with an RPI of 1-75 - Late-season performance in last eight games (strength and results) - Strength and result against conference opponents (regular-season and postseason) (15)

These are broad standards for selection and leave the potential for the selection committee to justify choosing any team they see as fitting "the best eligible student-athletes and teams of member institutions". Say for example that the selection committee is deciding between two teams for the final at-large bid in the NCAA Championship, Team A and Team B. Both teams have similar RPIs, similar schedules and compete in same conference. They tied in their head-to-head matchup. Team A has a record of 10-5-1 (6-0-0 in non-conference play and 4-5-1 in conference play), while Team B has a record of 10-5-1 (1-5-0 in

non-conference play and 9-0-1 in conference play). In this example, the selection committee could be justified in selecting Team A because of its superior non-conference schedule. However, the selection committee could also be justified in selecting Team B because of its superior in-conference schedule.

As shown in the example, the selection committee's power to arbitrarily decide merit can and has lead to controversy in practice (sources). Without a clear overarching system or metric, many teams can make their case for selection into the Championship. (The committee's deliberations are done in secret and their rationale is not officially revealed). However, it is commonly accepted in practice that the committee simply selects the twenty-six teams with the highest RPI, with perhaps one or two minor deviations, as detailed in this forum. Teams with high RPIs that fail to win their conference tournament can generally be confident that they will qualify for the NCAA Championship.

Thus, the RPI metric does not provide a wholly deterministic means of selection to the NCAA tournament. Its role and purpose in the selection process must be inferred, as the NCAA does not provide any formal explanation or justification for the usage of the RPI metric as a means of pairwise rankings. However, it is reasonable to assume for the purposes of this undergraduate thesis that the RPI provides a *baseline ranking system* from which teams can be adjusted according to the NCAA's arbitrary conception of the "best... teams of member institutions".

2.1.1 After Selection

After the 26 at-large teams are selection, the selection committee seeds the top 16 teams and gives them a first-round bye. The remaining teams are paired according to "geographical proximity" (13) in matchups that avoid potential intra-conference matches in the first two rounds of the championship.

It is important to note that there is *no official seeding procedure* for this process; the committee does not need to justify its choices based on any criteria. As noted by Chris Thomas on his site "RPI for Division I Women's Soccer," this is likely intentional. As he notes for the Women's Soccer Championship, which follows a similar procedure to the Men's Soccer Championship:

"An authoritative source has told me that it is "not unintentional" that the Manual does not state how the Committee is to seed teams. Thus although the Committee almost certainly considers the at large selection criteria in doing the seeding, the Committee is not bound by the criteria, nor is it foreclosed from giving weight to other considerations. **It also is worth noting that although the Manual says the Committee does not use coaches polls or any other outside polls or rankings "for selection purposes," it makes no mention of whether the Committee can use outside polls or rankings for seeding purposes.**"

Because of the lack of a clear procedure for the seeding process, the focus of this undergraduate thesis on RPI is motivated primarily by reducing bias in selection to the NCAA Men's Championship, rather than by the seeding process. A complete explanation is provided in the "RPI Usage Motivation" and "Claims

of RPI Bias” sections below.

After the selection committee chooses the bracket for the championship, games are played until a winner is determined.

After the conclusion of the season, coaches and athletics staff for the 212 NCAA D1 teams work to set their non-conference schedules by agreeing terms with other universities. It is important to note that results from the previous season and the final RPI rankings may impact how coaches select their non-conference matches for the following season. As RPI places a strong weight on strength-of-schedule, a team’s recent results may affect who chooses to play against them in non-conference play. Measuring the effect of a previous season’s RPI rankings on match selection for the following season is outside the scope of this paper and is an opportunity for future research.

2.2 Motivation for RPI

As mentioned in the previous section, the NCAA does not provide any formal explanation or justification for the usage of the RPI metric as a means of pairwise rankings for Men’s Soccer, besides acknowledging the fact that it is used to rank teams based on their strength of schedule. Therefore, the motivation behind the usage of the RPI metric must be inferred.

As mentioned before, this undergraduate thesis operates under the assumption that the RPI is intended to be a baseline ranking system according to the NCAA’s conception of who is “best.” The RPI then serves as an *approximation* of the NCAA selection committee’s idea of “merit.”

It is important to note that RPI is not a popular metric and is not used much in contexts outside of the NCAA. The coefficients of the metric are arbitrary and the metric does not have backing in statistical or mathematical theory for predictive accuracy (source). This makes the NCAA’s continued usage of the metric (despite some criticism) unusual and interesting. (NOTE: The NCAA did stop using the RPI metric in Men’s Basketball in 2018, replacing it with the NCAA Evaluation Tool (NET). Women’s Basketball switched to NET in 2020). Because of this, we can safely assume that the NCAA values features of the RPI metric for its specific context that do not apply to other contexts where the metric is not used. This can provide us with some intuition as to why the NCAA uses the RPI metric, but we can only guess.

Thus, this thesis does not seek to determine the rationale for the NCAA’s choice of the RPI metric as a baseline ranking system in comparison to other, more common pairwise-ranking systems such as the Bradley-Terry model (discussed in the “Procedure” section). Any conclusions on this topic are outside of the scope of this thesis. This thesis will operate only under the assumption and will draw conclusions from the idea that the NCAA uses the RPI as a quantitative baseline ranking system. The NCAA’s continued usage of RPI (despite criticism) is worth further research in a separate project. (A detailed mathematical breakdown of the RPI metric will be included in the “Procedure” section.)

It is now worth articulating why the NCAA would desire to have a quantitative baseline ranking system for teams in college soccer if the selection and seeding process is made arbitrarily by a human committee. Although this is the case, the language in the NCAA Men’s Soccer Prechampionship Manual valuing a team’s record and the intuition behind the existence of a championship clearly implies that the NCAA values teams which perform well on the field. It is almost trivial to state that the NCAA’s conception of merit, therefore, must include some notion of a team’s skill and performance on the soccer field.

However, a team’s demonstrated skill and performance is something that fluctuates over time. Therefore, it is worth asking *when* the NCAA most values a team’s performance with regard to selection to the Men’s Soccer Championship. Is the NCAA seeking to rank teams it feels will perform best in its championship, or is it seeking to quantify a team’s past performances over the season? The nature of RPI gives us an answer. RPI makes no effort to quantify time— all results from a season are equally weighted in the ranking. If the NCAA wanted to predict future performance in its quantitative baseline ranking system, one would expect it to factor time into its ranking system, but it has not.

Therefore, for this project, we can assume that the NCAA values a quantitative baseline ranking of teams by their past performances, rather than by their future potential. We can also assume that the NCAA uses RPI in order to provide this ranking system specifically in the context of selection for the NCAA D1 Men’s Soccer Championship. However, we will not make assumptions about why the NCAA uses RPI rather than other metrics for its desired quantitative baseline ranking of teams by their past performances.

2.3 Claims of RPI Bias

It has been claimed— especially in the context of Men’s Basketball— that RPI systematically overvalues teams from stronger conferences, who have more opportunities to play stronger teams and therefore increase their strength of schedule. In fact, the idea is so pervasive that it is included on the Wikipedia page for RPI— but without citation.

The mathematical intuition behind claims of bias is clear. Pairwise rankings should naturally account for strength of schedule, but it seems odd that the best quantitative baseline ranking of teams by their past performances would place clear majority of its weight on results outside of a team’s control. This also makes it so that losing can result in an increase in RPI (if against a good team) and so that winning can decrease a team’s RPI (if against a bad team) (source). This feels counterintuitive— a metric designed to evaluate performance should not reward a team when they play poorly.

2.3.1 RPI Bias test cases

For examples, we provide example cases of counterintuitive RPI adjustments, using unadjusted RPI.

For example, consider a league with 9 teams: A1, A2, A3, B1, B2, B3, C1, C2, C3. Suppose for this example that lower numbers always beat higher numbers (A1 always beats A2, ex.) and teams closer to the beginning of the alphabet always beat teams closer to the end (A1 always beats A2, ex.).

Suppose that teams have played 4 games: 2 against the other teams in their conference, as denoted by letter (A, B, or C) and 2 against the teams that share their number (1, 2 or 3). A photo of the matchups is provided in the Github repository.

The league table after these 4 matches is constructed as such.

RPI WP OWP OOWP

A1: 0.6953 1.0000 0.6250 0.5312
A2: 0.5977 0.7500 0.5625 0.5156
A3: 0.5000 0.5000 0.5000 0.5000
B1: 0.5977 0.7500 0.5625 0.5156
B2: 0.5000 0.5000 0.5000 0.5000
B3: 0.4023 0.2500 0.4375 0.4844
C1: 0.5000 0.5000 0.5000 0.5000
C2: 0.4023 0.2500 0.4375 0.4844
C3: 0.3047 0.0000 0.3750 0.4688

Here, A1 has an RPI of 0.6953 and C3 has an RPI of 0.3047.

Now suppose team A1 plays team C3 and wins. A1's winning percentage does not change (record still perfect). However, the opponent-based metrics drop because of C3's poor record and poor conference strength. (The same is true for C3, but the opponent metrics increase.)

The league table after this match is such:

RPI WP OWP OOWP

A1: 0.6312 1.0000 0.5000 0.5250
A2: 0.5898 0.7500 0.5625 0.4844
A3: 0.5000 0.5000 0.5000 0.5000
B1: 0.5898 0.7500 0.5625 0.4844
B2: 0.5000 0.5000 0.5000 0.5000
B3: 0.4102 0.2500 0.4375 0.5156
C1: 0.5000 0.5000 0.5000 0.5000
C2: 0.4102 0.2500 0.4375 0.5156
C3: 0.3688 0.0000 0.5000 0.4750

Thus A1 has an RPI of 0.6312 and C3 has an RPI of 0.3688. Thus A1 beat C3 and still dropped in RPI, and vice versa for C3. This is counterintuitive— we would expect a metric to reward teams for winning games and punish them for losing games. A metric as such introduces potentially problematic incentives. For example, A would be rewarded for canceling their game against C3, because they know it will hurt their RPI ranking, regardless of game result.

There are also scenarios where a team can lose to a team below them in the rankings and still move up.

Consider the scenario with teams A1, A2 and A3 in the A conference and B1, B2, and B3 in the B conference. Each team plays every team in its con-

ference once, with the lower number (closer to 1) always winning. Then, out of conference, A1 beat B3, A2 beat B2 and B1 beat A3.

The RPI after this scenario is as such:

RPI WP OWP OOWP

A1: 0.5370 1.0000 0.2222 0.7037

A2: 0.5278 0.6667 0.4444 0.5556

A3: 0.5093 0.0000 0.8889 0.2593

B1: 0.4907 1.0000 0.1111 0.7407

B2: 0.4722 0.3333 0.5556 0.4444

B3: 0.4630 0.0000 0.7778 0.2963

Now suppose A3 plays B1 again, and loses again. The RPI is then:

RPI WP OWP OOWP

A1: 0.5394 1.0000 0.2222 0.7130

A2: 0.5301 0.6667 0.4444 0.5648

A3: 0.5104 0.0000 0.9167 0.2083

B1: 0.4896 1.0000 0.0833 0.7917

B2: 0.4699 0.3333 0.5556 0.4352

B3: 0.4606 0.0000 0.7778 0.2870

Here, A3's RPI has improved from 0.5093 to 0.5104, despite their loss to a team below them in the RPI. Again, this is counterintuitive.

All of this context makes it clear that inter-conference RPI bias is an area for further study. Research has shown evidence of this kind of bias in NCAA Men's Basketball (11). However, the author notes that no academic research has sought to rigorously test the theory of inter-conference bias for NCAA D1 Men's Soccer from a formal mathematical standpoint.

It is worth indulging the theory that RPI bias towards teams from stronger conferences, if it exists, might be an intentional decision by the NCAA. Perhaps the NCAA seeks to reward stronger conferences for facilitating high levels of competition or for providing more TV revenue (as teams in larger conferences tend to have larger fan followings). However, this seems quite cynical, and past comments by committee chairs for other NCAA sports contradict this assumption. Former chairs of the NCAA Men's Basketball Selection Committee have indicated on multiple occasions that conferences are deemed as irrelevant in the selection process (sources). This thesis takes this claim, apply it to the D1 Men's Soccer selection process, and assumes it to be true.

Thus, it can be concluded that the NCAA would seek to avoid bias in its tournament selection process. As seen above, the mathematical intuition behind claims of inter-conference bias are reasonable and warrant investigation. Therefore, it is concluded that any findings on inter-conference bias in RPI (when compared to a "control" pairwise ranking metric) would be of value in the NCAA's assumed goal to have a quantitative baseline ranking of teams by their past performances which best informs the selection process for the D1 Men's Soccer Championship. This thesis seeks to provide those findings.

3 Procedure

This section seeks to provide information on the methods of testing inter-conference bias in RPI as well as the motivation behind their selection.

3.1 Data

Data for all matches during the 2024 NCAA D1 Men’s Soccer season has been scraped from the NCAA Men’s Soccer Scoreboard using this tool and Python code. Some of the code and the temp datasheets have been included in the project Github and will eventually be included or summarized in the Appendix.

A question is if more seasons are required for analysis. There are some factors that play into this:

- recent conference realignment
- perhaps the nature of the 2024 season predisposes itself to conference bias for one reason or another
- The purpose of the paper is to evaluate the bias of the metric in the present moment– i.e. if the metric is currently biased as is used by the selection committee. Thus, we should look to favor more recent seasons in our evaluation.
- The elimination of the overtime rule (likely) made ties less frequent (source), does this affect anything?

With these factors in mind, it is reasonable to look at more seasons than just the 2024 season– but not too many. Thus we will look at all seasons after the coronavirus pandemic, not including the abridged "2020" season. Thus, we will also scrape data for the 2021, 2022, and 2023 seasons. Findings for the 2024 season will take precedent but will be compared to the other three seasons for context and to see if there are major differences.

3.2 Models

In this subsection, the Rating Percentage Index and the Bradley-Terry Model will be explored.

3.2.1 RPI

3.2.2 Basic RPI

As mentioned above, the basic form of RPI is defined as follows:

$$RPI = 0.25 \times WP + 0.50 \times OWP + 0.25 \times OOWP \quad (1)$$

where WP = Winning Percentage,
OWP = Opponents’ Winning Percentage, and
OOWP = Opponents’ Opponents’ Winning Percentage.

Note that *ties count for 1/3 of a win in the winning percentage calculation.*

However, this metric does not factor in home-field advantage. Likely for this reason, the NCAA uses an Adjusted RPI ranking when ranking teams.

3.2.3 Adjusted RPI

The Adjusted RPI used by the NCAA for D1 Men’s Soccer rankings can be defined as follows:

$$RPI = 0.25 \times WP + 0.50 \times OWP + 0.25 \times OOWP + A \quad (2)$$

where A is a cumulative bonus or penalty for good wins/ties or bad losses/ties, determined by the RPI rank of the opponent. The full table of bonuses is outlined in the Prechampionship Manual on page 26.

The bonuses in the Adjusted RPI are only given to away teams, while the penalties are structure to hurt teams if they lose away (rather than at home). This is seemingly to reward teams for playing on the road and to factor home-field advantage into the ranking system.

Note that the penalty or bonus adjustment is calculated based on the separate initial RPI calculation. This ensures that the final Adjusted RPI is not dependent on the order in which bonuses are applied.

3.2.4 Other RPI Weights

A question to consider is whether other weights for the metrics in RPI should be considered. They will not be considered in this undergraduate thesis, but are an avenue for further research. An explanation as to why will be provided in due course.

3.2.5 Bradley-Terry

Although this thesis is designed to measure bias in RPI, it is necessary to have a standard quantitative pairwise ranking metric for comparison purposes. Because of this, RPI bias will be compared to that in the Bradley-Terry Model.

The author notes that Bradley-Terry is not the singular standard method for pairwise comparison. However, the Bradley-Terry model is commonly used in sports analytics and is a natural fit for this project (Rudi, personal communication) (Jamil 2010) (sources). A comparison of Bradley-Terry to other models such as Thurstone’s model would be justified but is not necessary given the ubiquity of Bradley-Terry model usage and the designated scope of this project.

3.2.6 Basic Bradley-Terry Model

The basic Bradley-Terry model gives the probability that team i beats team j as

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (3)$$

where λ_i and λ_j represent the respective strength parameters of teams i and j.

This provides us with a way to estimate the probability of a result given the strength coefficients of two teams. However, the NCAA championship selection process is concerned with team rankings rather than win probabilities. Therefore, we must estimate the distribution of the λ terms. These values will be derived using Bayesian inference, as recommended (Rudi, personal correspondence).

First, a prior distribution is set for the λ terms; from the standard normal distribution, for example. Then, given a set of game results, the likelihood function is

$$P(data | \lambda) = \prod_{i < j} \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right)^{w_{ij}} \left(\frac{\lambda_j}{\lambda_i + \lambda_j} \right)^{w_{ji}} \quad (4)$$

where w_{ij} represents the number of times team i beat team j. Then, using Bayes' Rule, the posterior distribution is:

$$P(\lambda | data) \propto P(data | \lambda)P(\lambda) \quad (5)$$

Where $P(\lambda)$ can be inferred from the prior.

The intuition and formulae behind Bayesian inference, as well as for the Markov Chain Monte Carlo (MCMC) techniques to derive the values, will be provided at a later date.

At the suggestion of Professor Rudi (personal communication) the Bradley-Terry model can be re-expressed in the form of logits, as shown. This has benefits for optimization efficiency and floating point stability (sources)

Here we will express the base Bradley-Terry model, but with π to represent strength parameters:

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j} \quad (6)$$

Then we apply a logit transformation $\lambda_i = \log(\pi_i)$, resulting in the following probability:

$$P(i \text{ beats } j) = \frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j}} \quad (7)$$

The logit transformation

$$\text{logit}(p) = \log \left(\frac{p}{1-p} \right)$$

then gives us

$$\text{logit}(P(i \text{ beats } j)) = \lambda_i - \lambda_j$$

which allows us to model the relationships between team strengths *linearly* on the log scale.

The likelihood function then becomes

$$P(data | \lambda) = L(\lambda) = \prod_{i < j} \left(\frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j}} \right)^{w_{ij}} \left(\frac{e^{\lambda_j}}{e^{\lambda_i} + e^{\lambda_j}} \right)^{w_{ji}} \quad (8)$$

As before, the posterior distribution can be expressed as

$$P(\lambda \mid data) \propto P(data \mid \lambda)P(\lambda) \quad (9)$$

Where $P(\lambda)$ can be inferred from the choice of prior.

Thus we are left to discuss the choice of prior.

The choice of prior distribution must reflect our beliefs about the strengths of the teams in our field of NCAA Men's Soccer teams. It is reasonable to suggest a *common prior* for all teams in the field, as suggested by Professor Rudi (personal communication). This is because RPI and the selection committee itself is solely focused on the results of the current season in the ranking and selection process. Thus, we will not determine priors based on this past information.

The question then becomes, what do we expect out of the distributions of strengths of our college soccer teams? How do we feel that team strengths should be grouped? Are most teams around the same level, or should they be grouped far apart? Will there be outliers, and how many? Do we expect some teams to be far and above better than others? And how does all of this fit into our desire to create a baseline model for testing ranking bias across conferences?

It is well known that soccer is one of the most luck-based sports (Professor Rudi, personal communication). With a small sample size of around 20 games for each team, it is naive to look to make strong assumptions about differences in team strengths. We should not expect to see many, if any, teams that are far better than the rest. Thus, it does not make sense to pick a prior that allows extreme values, like a Cauchy or Laplace prior.

In addition, the baseline nature of this model emphasizes practicality over accuracy or precision. We are measuring bias in the model *between conferences*, rather than predictive power. In fact, it might even make sense to have a weaker model, so that there are more predictive errors with which bias can be measured.

Thus, it is in this author's opinion that we should proceed with a normal prior, as such:

$$\theta_i \sim \mathcal{N}(\mu, \sigma^2)$$

This is a common choice. We will set $\mu = 0$ for all teams as a baseline value.

The question then becomes picking a value for σ . We want team distribution to be tight (not giving outliers too much sway, given the random nature of soccer) but also not too tight (as we do not want all teams ranked right next to each other). However, the exact prior is less important, because we are looking at rankings of teams rather than differences in team strengths. Because of this a value of $\sigma = 1$ is reasonable for this context, giving us a standard normal Gaussian prior

$$\theta_i \sim \mathcal{N}(0, 1)$$

Thus, $P(\lambda)$ can be obtained from the probability density function (PDF) of the standard normal distribution:

$$P(\lambda) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\lambda^2}{2}\right)$$

Thus we have a base Bradley-Terry model with a prior selected for the context of this thesis. However, there are necessary extensions this model that will be covered in the following sections.

3.2.7 Bradley-Terry-Davidson Model (BTDM)

The Bradley-Terry-Davidson model, as introduced by Davidson (1970), allows for the inclusion of ties in Bradley-Terry pairwise comparison.

It can be written as such: The probability that team i beats team j is

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j + \nu\sqrt{\lambda_i\lambda_j}}$$

and the probability that the two teams tie is

$$P(\text{draw}) = \frac{\nu\sqrt{\lambda_i\lambda_j}}{\lambda_i + \lambda_j + \nu\sqrt{\lambda_i\lambda_j}}$$

where ν is the new strength parameter for a tie.

The likelihood function then becomes:

$$P(\text{data} \mid \lambda, \nu) = \prod_{i < j} \left(\frac{\lambda_i}{\lambda_i + \lambda_j + \nu\sqrt{\lambda_i\lambda_j}} \right)^{w_{ij}} \left(\frac{\nu\sqrt{\lambda_i\lambda_j}}{\lambda_i + \lambda_j + \nu\sqrt{\lambda_i\lambda_j}} \right)^{t_{ij}} \left(\frac{\lambda_j}{\lambda_i + \lambda_j + \nu\sqrt{\lambda_i\lambda_j}} \right)^{w_{ji}} \quad (10)$$

and the posterior distribution is:

$$P(\lambda, \nu \mid D) \propto L(\lambda, \nu \mid D)P(\lambda)P(\nu)$$

for two priors $P(\lambda)$ and $P(\nu)$.

Again, we will use a log transformation. We have both

$$\theta_i = e^{\lambda_i}, \quad \eta = e^{\nu}$$

Leaving the following equations:

$$P(i \text{ beats } j) = \frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu}\sqrt{e^{\lambda_i}e^{\lambda_j}}}$$

$$P(\text{draw}) = \frac{e^{\nu}\sqrt{e^{\lambda_i}e^{\lambda_j}}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu}\sqrt{e^{\lambda_i}e^{\lambda_j}}}$$

and likelihood function

$$P(data \mid \lambda, \nu) = \prod_{i < j} \left(\frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ij}} \left(\frac{e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}} \right)^{t_{ij}} \left(\frac{e^{\lambda_j}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ji}}$$

with log

$$\log P(data \mid \lambda, \nu) = \sum_{i < j} w_{ij} \log \left(\frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}} \right) + t_{ij} \log \left(\frac{e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}} \right) + w_{ji} \log \left(\frac{e^{\lambda_j}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\lambda_i} e^{\lambda_j}}} \right)$$

And posterior

$$P(\lambda, \nu \mid D) \propto L(\lambda, \nu \mid D) P(\lambda) P(\nu)$$

for both priors.

For computational simplicity, and for many of the reasons listed in the previous subsection, we will use a standard normal prior for ν .

Thus

$$P(\nu) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{\nu^2}{2} \right)$$

3.2.8 BTDM with Home Weights

This thesis will use the "common" home-field-adjusted version of BTDM as outlined by Jamil (2010) for soccer purposes (22). Jamil outlines the formula on page 22 of his paper.

Now we have logit transformations

$$\theta_i = e^{\lambda_i}, \quad \eta = e^{\nu}, \quad \gamma = e^{\rho}$$

giving

$$P(i \text{ beats } j \mid i \text{ at home}) = \frac{e^{\lambda_i} e^{\rho}}{e^{\lambda_i} e^{\rho} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}}$$

$$P(draw \mid i \text{ at home}) = \frac{e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} e^{\rho} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}}$$

with likelihood function

$$P(data \mid \lambda, \nu, \rho) = \prod_{i < j} \left(\frac{e^{\lambda_i} e^{\rho_i}}{e^{\lambda_i} e^{\rho_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ij}^{home}} \left(\frac{e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} e^{\rho_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}} \right)^{t_{ij}^{home}} \\ \left(\frac{e^{\lambda_j} e^{\rho_j}}{e^{\lambda_j} e^{\rho_j} + e^{\lambda_i} + e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ji}^{home}} \left(\frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j} + e^{\nu} \sqrt{e^{\rho} e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ij}^{neutral}}$$

$$\left(\frac{e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{t_{ij}^{neutral}} \left(\frac{e^{\lambda_j}}{e^{\lambda_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ji}^{neutral}}$$

and log-likelihood function

$$\begin{aligned} \log P(\text{data} \mid \lambda, \nu, \rho) = & \sum_{i < j} \left(\frac{e^{\lambda_i} e^{\rho_i}}{e^{\lambda_i} e^{\rho_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ij}^{home}} + \left(\frac{e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} e^{\rho_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{t_{ij}^{home}} \\ & + \left(\frac{e^{\lambda_j} e^{\rho_j}}{e^{\lambda_j} e^{\rho_j} + e^{\lambda_i} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ji}^{home}} + \left(\frac{e^{\lambda_i}}{e^{\lambda_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ij}^{neutral}} \\ & + \left(\frac{e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}}{e^{\lambda_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{t_{ij}^{neutral}} + \left(\frac{e^{\lambda_j}}{e^{\lambda_i} + e^{\lambda_j} + e^\nu \sqrt{e^\rho e^{\lambda_i} e^{\lambda_j}}} \right)^{w_{ji}^{neutral}} \end{aligned}$$

and posteriors

$$P(\lambda, \nu, \rho \mid D) \propto L(\lambda, \nu, \rho \mid D) P(\lambda) P(\nu) P(\rho)$$

We will use our now-default choice of a standard normal prior for ρ , for simplicity and computational efficiency. (It is likely that $\mu > 0$, but to what degree is uncertain).

Thus

$$P(\rho) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\rho^2}{2}\right)$$

This is our final model. We will use this version of BTDM for comparison with Adjusted RPI.

3.2.9 Other Bradley-Terry Models

There are other examples of Bradley-Terry models that were considered as baseline ranking systems for this thesis but which were eventually rejected. The rationale behind these decisions is quickly outlined here and will be elaborated on in the future.

Bradley-Terry-Davidson model which accounts for score: Score is not something that the selection committee appears to consider. For our base model, we are not looking to factor in metrics that are outside the committee's selection criteria in a metric. We feel that, if score was a desirable measure for ranking, it would be included in the RPI formula. Thus, we are ignoring score and taking results at face value with our version of the Bradley-Terry-Davidson model.

3.3 Testing

3.3.1 Bias Measurement

In order to measure what we call "bias" across conference, we must have some idea of deviation from a "true" ranking. Bias cannot be evaluated in a vacuum—it must be relative to a ground truth, or at least to another metric.

To make this evident, we provide a simple example. Suppose there are two teams that are, in ground truth, equally weak. They have both lost all of their games. However, Team A has played stronger opponents than Team B. It is *necessary* that Team A be ranked higher than Team B in any reasonable ranking system. Otherwise, we will just be evaluating teams based on their record—penalizing teams for playing difficult games and incentivizing teams scheduling unfair matchups against opponents much worse than them. This, clearly, is no way to approach a ranking system—it discentivizes true competition.

Thus Team A should be ranked above Team B. However, without access to any value of "ground truth" strength, the extent to which Team A is ranked above Team B is open to interpretation.

This is because the true data generation method—the results of matches—is really messy. In the real world, we don't know the true process with which matches are decided, and what is just random noise. The best we can do with real data is compare a ranking method to a similar metric which is also necessarily imperfect.

However, we can also perform simulations which allow us to have access to the ground truth values of team strength as well as access to the data generation process. We can then compare our ranking methods to this objective idea of what the rankings *should* be from the simulations.

We will use two methods for evaluating bias across conferences. The first is by comparing the Average Difference from True Ranking (ADTR). This means evaluating conference bias by averaging the difference in a team's rank compared to what their rank actually should be.

A simple example is provided with the photo in the Github. In the photo, there are four teams: A1, A2, B1 and B2. Suppose that the column on the left is the "ground truth" ranking of the teams, in our simulation. The column on the right then represents the projected ranking of the teams using our metric. In this ranking, A2 is ranked above B1, even though it should be ranked below B1. A2 then overperforms its ground truth ranking by 1 point, while B1 underperforms by 1 point. The ATDR would then be $1/2 = 0.5$ points for Conference A and $-1/2 = -0.5$ for Conference B.

The second metric will be an application of the Spearman's rank coefficient for conference comparison purposes.

The general equation is:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks of the i th item in the two lists and n is the total number of ranked items.

We will be modifying Spearman's Rank Coefficient to be calculated for each conference. The coefficient becomes for each conference:

$$\rho_C = 1 - \frac{6 \sum (d_{Ci})^2}{n_C(n_C^2 - 1)}$$

where d_{Ci} is the difference between the ranks of the i th team in the two conferences and n_C is the total number of teams in the conference.

A ranking system that doesn't have underlying bias compared to ground truth rankings by conference should have a distribution of ρ_C that is completely random. We want to look at patterns in Spearman's Rank across many simulations to see, on average if there is a noticeable bias that correlates with conference strength.

Thus, we will calculate Spearman's Rank for both RPI and for BTDM and look to compare the two. We will also look at the variance and the Fischer-Pearson skewness of both distributions of Spearman's Rank to better quantify how bias is distributed across conferences.

After simulation, we will look at real data from the past 4 years. Note that we will no longer have access to a ground truth distribution of teams. this is important— *at the end of the day— there's no true way to fully assess "bias" without an idea of ground truths.*

This is okay. However, we must change our comparison. We can quantify and compare the correlation between RPI and conference with BTDM and conference, but you can't quantify whether this is actually "biased" or "unfair".

The intention is that, if the BTDM doesn't show bias in simulations and RPI does, then we can *assume* BTDM are closed "ground truth" rankings from which RPI can be compared to it and we can see if RPI is more biased for it across conferences. This will then lead to a couple of questions. If we had used the BTDM last year, what would it have looked like? And how would the BTDM have affected ranks by conference?

However, we will not focus on that for now. We will start with the simulation, then look at what the simulations are telling us.

3.3.2 Simulation Procedure

Evaluation of bias will start with simulations. A key question to first consider is whether to create synthetic matches. The author has been advised against by Professor Rudi (personal communication). Further explanation as to why will be added in due course.

We will run a series of basic simulations on the data, to serve as useful test cases to observe RPI and BTDM bias across conferences in different scenarios. It is important to look at RPI and BTDM in various constructed situations, for the following purposes, among others: - What happens if teams are spread out in strength, or more similar? - What happens if results are more luck based, or more deterministic? - What if there is no differences in team strength by conference, or there are differences by conference?

Here is a list of a basic simulations and the motivations behind them:

- Deterministic results (a hidden ranking of all of the D1 teams is created, and all results follow those rankings). This is a useful base case for comparing ranking systems to start with. We should not expect to see a pattern of bias in this simulation because there is no conference strength factor.

- All teams have equal probability of winning, tying and losing in every game. This is another useful base case, where results are completely determined by random luck. Again, we should not expect to see a pattern of bias in this simulation because there is no conference strength factor.

- An underlying conference ranking is determined, the probability of a team winning is then (opponents conference) / (my conference + opponents conference), with no chance for ties. This simulation allows for a simple method of determining results quite heavily from conference rank, but not completely. This is a good prelude to the more "realistic" logistic simulations in the next two simulation methods.

- All teams are given a strength parameter from the uniform distribution (0, 1) that is consistent within conferences; the probability of a team winning will be then logistic: $\exp(\text{their strength}) / (\exp(\text{their strength}) + \exp(\text{their opponent's strength}) + \sqrt{\exp(\text{their strength}) * \exp(\text{their opponent's strength})})$. (A tie has the odds $\sqrt{\exp(\text{their strength}) * \exp(\text{their opponent's strength})} / (\exp(\text{their strength}) + \exp(\text{their opponent's strength}) + \sqrt{\exp(\text{their strength}) * \exp(\text{their opponent's strength})})$), this is equivalent to a Bradley-Terry-Davidson tie coefficient of 1).

This simulation allows us to evaluate a semi-random process in which conference play a role in how results are determined. With a non-deterministic random distribution and the chance for ties, this should hopefully be a more "realistic" simulation of results, yet without any individual team strength factors. This is a logistic simulation of results, a standard choice in scenarios like this one (sources). Because results are simulated in the same method that Bradley-Terry-Davidson is calculated, we expect (though are not certain) the BTDM model to do well here. The main question is how RPI will do.

- All teams are given a strength parameter from the uniform distribution (0, 1) PLUS an additional parameter from the uniform distribution (0, 1) for conference strength; the probability of a team winning will be $\exp(\text{their strength}) / (\exp(\text{their strength}) + \exp(\text{their opponent's strength}) + \sqrt{\exp(\text{their strength}) * \exp(\text{their opponent's strength})})$. (A tie has the odds $\sqrt{\exp(\text{their strength}) * \exp(\text{their opponent's strength})} / (\exp(\text{their strength}) + \exp(\text{their opponent's strength}) + \sqrt{\exp(\text{their strength}) * \exp(\text{their opponent's strength})})$), this is equivalent to a Bradley-Terry-Davidson tie coefficient of 1).

This simulation allows us to evaluate a semi-random process in which both individual team strength as well as conference plays some, but not the entire role in how results are determined. Because results are simulated in the same method that Bradley-Terry-Davidson is calculated, we expect (though are not certain) the BTDM model to do well here. The main question is how RPI will do.

We will run these simulations many times separately on the matches for the 2024 season. We will only be looking at the 2024 season to reduce comparisons

across seasons when conference alignment has changed. For example, the Pac-12 conference existed in 2023 but not 2024 (source), so we might have different ranking results across seasons if there is a large variation in what the conference matchups are.

The 2024 season might not be a perfect proxy for NCAA matchups (for example, certain matchups might consistently hold strong weight in cross-conference comparison), but is a reasonable approach given the high variability in conferences across seasons in the NCAA. (For the portion of this thesis which looks at real data, we will look at seasons conferences for 2021-2024.)

Bias evaluation metrics will be averaged over every simulation, as mentioned in the section above. Simulations will be conducted using Python.

3.3.3 Analysis of Real Data

TBD— after the conclusion of the simulation phase of this undergraduate thesis.

4 Results

4.1 Simulation